

Projects and Demos Inventory

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Embedded Common C and C++

- Static heap allocators with dynamic allocation semantics for advanced data structures (open address hash table, RB tree, graph)
- Compatible with `std::c++`
- C++14 conforming `steve::vector` with static allocator

STM32CubeIDE

- Interrupt bare metal "multi-threading" with `std::atomics`
- System Graph
- STM32CubeMX pinout configuration
- Semihosting

Keil uVision with STM32CubeMX

- STM32CubeMX pinout config integrated
- Interrupt bare metal "multi-threading" with C11 atomics

VSCode

- Rust embedded bare metal state captures
- Semihosting
- Cortex Quick Start for STM32F401RE

Visual Studio

- Raspberry Pi remote cross compile and debugging

MLibs

- Embedded automation suite
- Semihosting

Raspberry Pi Assembly

- Native assembly development (sorts and other routines)

Algorithms C++ General

- Automata
- Graph
- Linked List
- Stack
- Queue
- Heap Sort
- Merge Sort
- Quick Sort
- Open address hash table with cluster diagnostics
- Red Black Tree full implementation
- C++14 conforming `steve::vector` with static allocator
- Multithreaded, SIMD matrix multiplication
- Add bits
- Multidrive whole system file cataloging
- Rust code generator simple "wizard"
- FreeBSD/Linux Tree and List macros

Algorithms Rust

- Kadane
- Linked List
- Binary Tree
- Add Bits
- Quick Sort
- Bounce Filter
- Two Pointers
- Is Anagram
- Valid Parens, Brackets, and Braces
- Binary Search
- Automata (Works with inorder vowels, automata parity, automata reject three)
- Find in Files
- Insertion Sort
- String Permute
- Dekker's Algorithm (lock)
- Equal to Subarray
- Jumping Numbers
- Interleaving
- Max Index Span
- Sum of Products
- Non-Repeating
- Rect Intersect
- Remove Duplicates
- Rotate Grid
- Tic-tac-toe
- Zeros to End